

Predicting Used Car Prices with Machine Learning: A Comparative Study of Models and Preprocessing Strategies

Yifan Wu (1618332) Xinkai Zhang (1619423) Zhenyu Zhang (1255206) Yiming Zhang (1681020)

Abstract

Used car price prediction is a practically important regression task that requires careful handling of heterogeneous tabular data. In this paper, we investigate the effectiveness of machine learning methods for predicting second-hand vehicle prices using the UK Used Car Dataset, comprising approximately 100,000 listings across 11 manufacturers. We address four research questions concerning prediction accuracy, model selection, the impact of preprocessing strategies, and feature importance. We compare five models spanning high-bias to low-bias: Linear Regression, Ridge Regression, K-Nearest Neighbours, Random Forest, and a Multi-Layer Perceptron. Our experiments demonstrate that ensemble tree-based methods and neural networks substantially outperform linear baselines (RMSE improvement of X%), that log-transforming the skewed price target and applying group-median imputation are the most impactful preprocessing choices, and that vehicle age and annual mileage are the dominant price predictors.

1 Introduction

The second-hand car market is one of the largest consumer goods markets globally, with millions of transactions per year. Accurately estimating a fair market price benefits both buyers—who risk overpaying—and sellers—who risk underpricing their vehicles. Traditional valuation relies on domain expertise and lookup tables, but these approaches fail to capture the complex, non-linear interactions among vehicle attributes such as brand, age, mileage, fuel type, and transmission. Machine learning offers a data-driven alternative that can model these interactions directly from historical listing data (Gegic et al., 2019).

This project investigates machine learning methods for used car price prediction using the UK Used Car Dataset (Desai, 2021), a publicly available collection of ~108,000 listings scraped from a major

UK automotive marketplace. The dataset provides a rich combination of numerical and categorical features, real-world missing values, and a heavily right-skewed continuous target (price in GBP), making it well-suited for exploring both modelling and preprocessing questions.

Research Questions. We address four specific research questions (RQs):

- RQ1** To what extent can machine learning models accurately predict used car prices from available features?
- RQ2** How do models spanning the bias–variance spectrum (linear models, instance-based, tree-based, and neural networks) compare in predictive performance and generalisation?
- RQ3** How do different data preprocessing strategies—target transformation, missing value imputation, and feature scaling—affect model performance?
- RQ4** Which features contribute most significantly to price prediction, and how can their importance be interpreted?

RQ1 establishes a baseline understanding of task difficulty. RQ2 provides a systematic model comparison linked to the bias–variance trade-off. RQ3 directly examines the downstream impact of preprocessing decisions, with particular relevance to distance-sensitive models such as KNN. RQ4 connects model behaviour to domain knowledge about the used car market.

The remainder of the paper is structured as follows: Section 2 reviews related work; Section 3 describes our dataset, preprocessing pipeline, models, and evaluation setup; Section 4 presents experimental results; Section 5 analyses findings; and Section 6 concludes.

2 Literature Review

Price prediction for used vehicles has attracted considerable attention in the machine learning literature. [Pudaruth \(2014\)](#) provided an early comparative study applying Naive Bayes, Decision Trees, and Neural Networks to a small Mauritian dataset, finding that tree-based methods outperform Naive Bayes due to the non-Gaussian nature of automotive features. [Gegic et al. \(2019\)](#) extended this to a larger European dataset using gradient-boosted trees and Support Vector Regression, demonstrating that ensemble methods consistently outperform linear baselines, with mileage and vehicle age emerging as the two most predictive features—a finding we replicate and contextualise in Section 4. [Shinde and Kotecha \(2018\)](#) proposed a KNN-based system for used car valuation, showing that KNN achieves competitive accuracy on datasets with strong local price clusters (e.g. vehicles of the same model and year), but degrades significantly without feature scaling—motivating our investigation of preprocessing effects in RQ3. [Listiani \(2009\)](#) applied SVR to German car listings, finding that RBF kernels outperform linear kernels, and that log-transforming the price target substantially improved RMSE.

Neural network approaches have gained traction with the availability of larger datasets. [Goodfellow et al. \(2016\)](#) provide the theoretical basis for multi-layer perceptrons as universal approximators, supporting their use for capturing non-linear feature interactions. [Breiman \(2001\)](#) introduced Random Forests, which have become a strong baseline for tabular regression due to their robustness to irrelevant features and insensitivity to feature scale.

Our work contributes to this literature by: (1) using a substantially larger and more diverse dataset than prior studies; (2) conducting a controlled comparison across five model families; and (3) performing a systematic ablation of preprocessing strategies, which prior work treats as fixed.

3 Method

3.1 Dataset

We use the **UK Used Car Dataset** ([Desai, 2021](#)), a publicly available collection of used vehicle listings scraped from a major UK online automotive marketplace. The raw dataset comprises 108,540 listings distributed across 11 CSV files, one per manufacturer: Audi, BMW, Ford (two subsets), Mercedes (two subsets), Hyundai, Skoda, Toyota,

Vauxhall, and Volkswagen. All files share a common nine-column schema: model, year, price (GBP), transmission, mileage, fuelType, tax (annual road tax in GBP), mpg (miles per gallon), and engineSize (litres). Two additional “unclean” files (Ford Focus, Mercedes C-Class) are included as raw scrape outputs with malformed price strings and split mileage columns; these are used to illustrate real-world data quality challenges but are not included in the main training corpus.

Properties. Table 1 summarises the key properties of the combined dataset after loading. The target variable price is continuous and exhibits severe right-skew (skewness = 2.29), with a median of £14,698 and values ranging from £450 to £159,999. Two features—tax (13.1% missing) and mpg (8.6% missing)—contain substantial proportions of missing values arising from incomplete scraping. The dataset contains 9 brands after merging the two Ford and two Mercedes subsets, and 195 distinct car models. Linear separability between price strata is low given the continuous target, but a moderate negative correlation with mileage ($r = -0.47$) and positive correlation with year ($r = 0.52$) provide a clear signal for regression.

Property	Value
Total listings (raw)	108,540
Listings after cleaning	~100,000
Number of brands	9
Number of models	195
Numerical features	6 (year, price, mileage, tax, mpg, engineSize)
Categorical features	3 (model, transmission, fuelType)
Target variable	price (continuous, GBP)
Target skewness (raw)	2.29
Target skewness (log)	-0.13
Missing values (tax)	13.1%
Missing values (mpg)	8.6%

Table 1: Summary statistics of the UK Used Car Dataset before and after cleaning. Listing counts are approximate post-cleaning due to duplicate and outlier removal.

3.2 Data Preprocessing

All preprocessing is implemented in Python using pandas and scikit-learn, and is fully reproducible via Notebook 01 in the accompanying code repository. The pipeline proceeds through five stages.

Stage 1: Schema Standardisation. Each per-brand CSV is loaded and filtered to the nine standard columns, discarding any extraneous columns

(e.g. the `tax(£)` column present in the Hyundai file due to a scraping inconsistency). All files are concatenated into a single DataFrame and a brand column is appended.

Stage 2: Validity Filtering. Four categories of physically implausible values are identified and treated as missing rather than deleted, preserving sample size:

- `engineSize = 0`: no internal combustion vehicle has zero displacement; likely a data-entry error.
- `mileage = 0`: used car listings with zero recorded mileage are implausible placeholders.
- `price < 100`: below any realistic transaction threshold.
- `year ∉ [1990, 2023]`: values of 1970 and 2060 represent obvious entry errors.

Stage 3: Duplicate Removal. Exact duplicate rows (identical across all nine columns) are removed.

Stage 4: Outlier Removal. We apply the IQR method with a conservative factor of $3\times$ to price, mileage, and tax. A row is removed if its value falls below $Q_1 - 3 \cdot \text{IQR}$ or above $Q_3 + 3 \cdot \text{IQR}$. This factor retains genuine high-value luxury cars and high-mileage fleet vehicles while discarding clear data errors. The resulting price bounds are approximately £[1,229, 46,440].

Stage 5: Category Consolidation. Five fuel type labels and four transmission labels in the raw data include rare entries (e.g. *Other*, *Electric* for a small fraction of the dataset). We retain only `{Petrol, Diesel, Hybrid, Electric}` and `{Manual, Automatic, Semi-Auto}`, mapping all other values to NaN for subsequent imputation.

3.3 Missing Value Handling

Table 2 shows the missing value profile after validity filtering. We use **group-median imputation** for all numerical columns and **within-brand mode imputation** for categorical columns. Group-median imputation is preferred over global median because feature distributions are strongly stratified by brand and fuel type: for example, diesel engines have systematically higher displacement and different MPG profiles than petrol engines.

Feature	Imputation Strategy	Missing %
tax	Median within (year, engine bin)	~13%
mpg	Median within (fuelType, engine bin)	~9%
engineSize	Median within (brand, fuelType)	<1%
mileage	Median within (year, brand)	<1%
fuelType	Mode within brand	<1%
transmission	Mode within brand	<1%

Table 2: Missing value rates (post-cleaning) and imputation strategies. Engine bin refers to a coarse discretisation of engineSize into five intervals.

3.4 Feature Engineering

Beyond the nine raw features, we construct four derived features motivated by domain knowledge of vehicle depreciation:

- **car_age** = $2024 - \text{year}$: vehicle depreciation is primarily a function of age rather than the registration year in isolation.
- **mileage_per_year** = $\text{mileage} / (\text{car_age} + 1)$: normalises mileage by age, distinguishing a high-mileage new car from a low-mileage old one. A car driven 20,000 miles in one year signals different usage patterns than one driven 100,000 miles over ten years.
- **is_luxury** $\in \{0, 1\}$: binary indicator for premium brands (Audi, BMW, Mercedes), which command a price premium independent of other features.
- **log_price** = $\log(1 + \text{price})$: log-transformation of the target to reduce right-skew (skewness from 2.29 to -0.13). All models are trained to predict `log_price`; results are exponentiated for interpretation.

3.5 Categorical Encoding

We apply three encoding strategies tailored to the cardinality and ordinality of each categorical feature:

- **Label encoding** for transmission (3 levels): `Manual` $\rightarrow 0$, `Semi-Auto` $\rightarrow 1$, `Automatic` $\rightarrow 2$. The ordering reflects increasing automation level, making this a reasonable ordinal treatment.
- **One-hot encoding (OHE)** for fuelType (4 levels): avoids imposing a false ordinal relationship. This produces four binary indicator columns (`fuel_Petrol`, `fuel_Diesel`, `fuel_Hybrid`, `fuel_Electric`).

- **Target encoding** for brand and model: each category is replaced by the mean `log_price` of that group. Target encoding is preferred over OHE for these high-cardinality features (9 brands, 195 models) because it produces a compact, informative representation without the curse of dimensionality. In the modelling notebooks, target encoding statistics are computed on the training fold only to prevent label leakage.

3.6 Feature Scaling

The final modelling feature set comprises 14 features (Table 3). We prepare two scaled versions for experiments related to RQ3:

- **StandardScaler**: zero mean, unit variance. Standard choice for gradient-based models and regularised linear models.
- **MinMaxScaler**: maps each feature to $[0, 1]$. Used to study the effect of bounded vs. standardised inputs on KNN and neural network performance.

Tree-based models (Random Forest) are scale-invariant and always trained on unscaled features.

Feature	Type	Source
<code>car_age</code>	Numerical	Engineered
<code>mileage</code>	Numerical	Raw
<code>mileage_per_year</code>	Numerical	Engineered
<code>tax</code>	Numerical	Raw
<code>mpg</code>	Numerical	Raw
<code>engineSize</code>	Numerical	Raw
<code>transmission_enc</code>	Ordinal	Label-encoded
<code>is_luxury</code>	Binary	Engineered
<code>brand_mean_price</code>	Numerical	Target-encoded
<code>model_mean_price</code>	Numerical	Target-encoded
<code>fuel_Diesel</code>	Binary	OHE
<code>fuel_Electric</code>	Binary	OHE
<code>fuel_Hybrid</code>	Binary	OHE
<code>fuel_Petrol</code>	Binary	OHE
Target: <code>log_price</code>	Numerical	Engineered

Table 3: Final modelling feature set (14 features) used as input to all models.

3.7 Validation Setup

To enable fair comparison across all models and preprocessing variants, we adopt a unified validation protocol. The dataset is split into a **training set** (80%, stratified by brand) and a held-out **test set** (20%) using a fixed random seed (42). The test set is used only for final evaluation and is never seen during model selection or hyperparameter tuning.

Hyperparameter tuning is conducted using **5-fold cross-validation** on the training set. The same folds are used for all models to ensure that observed performance differences are not attributable to fold assignment. Target encoding statistics (for brand and model) are recomputed within each fold to prevent label leakage.

3.8 Models and Hyperparameters

We evaluate five models covering the bias–variance spectrum from high-bias/low-variance to low-bias/high-variance:

1. **Ridge Regression** — linear model with L2 regularisation; $\alpha \in \{0.01, 0.1, 1, 10, 100\}$.
2. **K-Nearest Neighbours (KNN)** — instance-based; $k \in \{3, 5, 10, 20, 50\}$; distance weighting.
3. **Support Vector Regression (SVR)** — kernel SVM; RBF kernel; $C \in \{0.1, 1, 10\}$, $\varepsilon \in \{0.01, 0.1\}$.
4. **Random Forest** — ensemble of 100–500 decision trees; `max_depth` $\in \{10, 20, \text{None}\}$; `min_samples_leaf` $\in \{1, 5\}$.
5. **Multi-Layer Perceptron (MLP)** — two hidden layers; `width` $\in \{64, 128\}$; `dropout` $\in \{0.0, 0.2\}$; Adam optimiser; early stopping.

3.9 Evaluation Metrics

All models are evaluated on the same three metrics, computed on original-scale predictions (after exponentiating `log_price` predictions):

- **RMSE** (Root Mean Squared Error): penalises large errors more heavily; primary metric.
- **MAE** (Mean Absolute Error): interpretable as average absolute price deviation in GBP.
- **R²** (Coefficient of Determination): proportion of price variance explained by the model.

For preprocessing comparisons (RQ3), we additionally report the cross-validation RMSE and its standard deviation to assess stability across folds.

4 Results

[Results to be completed after model training and evaluation.]

5 Discussion

[Discussion to be completed after results analysis.]

6 Conclusion

[Conclusion to be completed after all sections are written.]

Generative AI Usage

Claude (Anthropic) was used to assist with: (1) brainstorming research questions during the project planning phase; and (2) suggesting the initial project directory structure. No generative AI tools were used to write any part of this report, generate any dataset, or produce any experimental results. All analysis, code, and writing were produced by the authors.

References

- Leo Breiman. 2001. Random forests. *Machine Learning*, 45(1):5–32.
- Aditya Desai. 2021. UK Used Car Dataset. <https://www.kaggle.com/datasets/adityadesai13/used-car-dataset-ford-and-mercedes>. Accessed: May 2026.
- Enis Gegic, Bakir Isakovic, Dino Keco, Zerina Masetic, and Jasmin Kevric. 2019. Car price prediction using machine learning techniques. In *TEM Journal*, volume 8, pages 113–118.
- Ian Goodfellow, Yoshua Bengio, and Aaron Courville. 2016. Deep learning.
- Mandalina Listiani. 2009. Support vector regression analysis for price predictions in the german car market. *Thesis, Hamburg University*.
- Sameerchand Pudaruth. 2014. Predicting the price of used cars using machine learning techniques. *International Journal of Information & Computation Technology*, 4(7):753–764.
- Sneha Shinde and Ketan Kotecha. 2018. Used car price prediction using K-Nearest Neighbor based model. *International Journal of Innovative Research in Computer and Communication Engineering*, 6(5).